

Worksheet 2: Decimals – 2nd ESO

1. Theory

Decimals allow us to represent numbers more precisely than whole numbers. Frequencies on the FM radio dial are decimal numbers because they represent values between whole numbers (e.g., 87.5 MHz).

Decimal place value

Each digit after the decimal point represents tenths, hundredths, thousandths, etc. This helps define exact frequencies.

Rounding and precision

In electronics, rounding decimals can affect tuning accuracy.

2. Exercises

Rounding

Round the following frequencies to two decimal places:

- 99.876 MHz
- 101.2345 MHz
- 87.599 MHz

Ordering

Order the frequencies from smallest to largest:

- 90.12 MHz
- 90.3 MHz
- 90.03 MHz
- 89.99 MHz

Conversion: MHz to kHz

- 88.7 MHz = ? kHz
- 102.35 MHz = ? kHz

3. Competence-based problems

Problem 1

If a radio tuner moves in steps of 0.05 MHz, how many steps does it take to go from 88.0 MHz to 108.0 MHz?

Problem 2

Calculate the difference in kHz between 98.25 MHz and 98.5 MHz.

Solutions

Rounding:

- $99.876 \rightarrow 99.88 \text{ MHz}$
- $101.2345 \rightarrow 101.23 \text{ MHz}$
- $87.599 \rightarrow 87.60 \text{ MHz}$

Ordering:

$89.99 < 90.03 < 90.12 < 90.3 \text{ MHz}$

Conversion:

- $88.7 \text{ MHz} = 88,700 \text{ kHz}$
- $102.35 \text{ MHz} = 102,350 \text{ kHz}$

Problems:

- 1. Number of steps = $(108.0 - 88.0) / 0.05 = 400 \text{ steps}$
 - 2. Difference = $(98.5 - 98.25) \text{ MHz} = 0.25 \text{ MHz} = 250 \text{ kHz}$
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